

EE 695: Sliding Mode Control

Course Code: EE 695

Course Name: Sliding Mode Control

L-T-P-C: 3-0-0-3

Prerequisites: Linear Control Systems Theory

Intended for: B. Tech Final year, M. Tech. and Ph. D. students

Mutual Exclusion: None

Approval: 61st BoA (to be reported to Senate)

Preamble

Slide mode control is a simple and yet robust control technique. In case of sliding mode control, the system states are made to confine to a selected subset of the state space so as to achieve some desirable dynamics. Traditionally, a relay-based control has been used for this purpose and had its roots in the variable structure system philosophy. Developed in the erstwhile Soviet Union, the concept was pioneered by Vadim Utkin. Sliding mode control has numerous applications in defence, reactor control, robotics, etc. and it is one of the fastest growing fields.

The course will start from the fundamentals of sliding mode control. The history of sliding mode control will be discussed next. The basic theory of sliding mode control design will be discussed such as sliding surface design and control design. Much emphasis will be given to discrete time sliding mode control as the control is implemented digitally. After that the advanced topics on sliding mode control will be discussed. The sliding mode observer and integral sliding mode control will also be discussed. Finally, higher order sliding mode control and event triggered sliding mode control will be discussed.

After going through this course, the learners will be able to apply continuous-time and discrete-time sliding mode controllers and observers to practical systems, as alternative to other control techniques, besides further advanced learning and pursuing theoretical research in the area.

Course Modules

- Introduction to sliding mode, History and origin of sliding mode, Continuous sliding mode, Filippov solution, Sliding surface design, Equivalent control, unit vector approach, sliding mode control for MIMO system. (8 Lectures)
- Discrete time sliding mode control (DSM), Need for discrete time sliding mode and definition of DSM, Quasi sliding mode (QSM), QSM band, Reaching law based DSM, Switching and non-switching based DSM, Multi-rate output feedback based DSM. (8 Lectures)
- Luenberger observer, Reduced order observer, Sliding mode observer, Modified sliding mode observer, Integral sliding mode control (ISM), Composite nonlinear feedback control (CNF), CNF-ISM control design, Sliding mode control with actuator saturation. Tracking control for non-minimum phase system. (10 Lectures)
- Higher order sliding mode control, Twisting and Super-twisting control, Higher order sliding mode based observer and differentiator, Continuous integral sliding mode control, Implementation of super-twisting control by higher order finite time observer (10 Lectures)
- Event triggered control, Event triggered sliding mode control, Discrete time event triggered sliding mode control, Periodic event triggered sliding mode control. (4 Lectures)

Laboratory/practical/tutorial Modules

None

Text Books

1. Christopher Edwards and Sarah Spurgeon, *Sliding Mode Control: Theory and Applications*, CRC Press, London, UK, 1998.
2. Y. Shtessel, C. Edwards, L. Fridman and A. Levant, *Sliding mode control and Observation*, Birkhauser, (a division of Springer, New York.) 2013.
3. B. Bandyopadhyay and S. Janardhanan, *Discrete-time sliding mode control: A multi-rate Approach*, LNCIS, Springer Verlag, Germany, 2006.